

HTTP Server Basic

Description: Write a Python program using the socket module that creates a basic HTTP server returning a Hello World HTML page.

Your task is to implement the following functions:

1. `create_server()` -> `socket.socket`: This function creates a TCP socket, sets `SO_REUSEADDR`, binds to `('localhost', 8080)`, listens with backlog 5, and returns the socket.
2. `generate_response()` -> `bytes`: This function returns a complete HTTP 200 response with Content-Type: `text/html` and a body of `'<html><body><h1>Hello, World!</h1></body></html>'`.
3. `serve()` -> `None`: This function runs the server using `select`, accepts connections, sends the response from `generate_response()`, and handles `KeyboardInterrupt` to close the server.

Input: None.

Output (without unit test): The server runs and handles connections.

Output (with unit test):

```
Testing create_server ...
bind called with: call(('localhost', 8080))
listen called with: call(5)
Testing generate_response ...
test attribute passed: b'Hello, World!' is in b'HTTP/1.1 200 OK\r\nContent-Type:
text/html\r\nContent-Length: 48\r\n\r\n<html><body><h1>Hello, World!</h1></body>
</html>'
Testing serve ...
accept called with: call()
test attribute passed: call(('localhost', 8080)) is equal to call(('localhost',
8080))
test attribute listen passed: True is True
test attribute accept passed: True is True
test attribute close passed: True is True
```